Python Basics Worksheet: Solve It, Then Ask the AI

How to use this worksheet

You will solve 10 exercises that get harder step by step. Each exercise has three parts:

1. **Theory box** — the Python you need, in a few lines.
2. **Solve it alone** — write and run your own code first. No AI yet. A wrong first try is fine; that is how programming works.
3. **Compare with GenAI** — copy the prompt into ChatGPT, Claude, Gemini or Copilot. Send only the exercise, never your own code. Then compare the AI's solution with yours using the questions below.

The same three questions apply to every comparison:

- Does the AI's code give the same result as yours? Test both with at least two inputs.
- What does the AI do differently? Is it shorter, longer, clearer, or does it use something you have not learned yet? If there is something you don't understand, ask!
- Did the AI make any mistake or break a rule from the exercise?

The prompt template (already filled in for each exercise below):

```
I am a beginner in programming learning the basics of Python.
I only know: [CONCEPTS].

Exercise: [EXERCISE TEXT]

Write a beginner-level solution using only the concepts above. Explain how it works
in 3-4 short sentences.
```

Why the line "I only know…"? Without it, the AI often uses tools you have not seen yet. Try removing it once and see what happens.

Part 1 — Variables and data types

Exercise 1 — Introduce yourself

> ***Theory box: variables and print***
>
> *A variable is a named box that stores a value:* `name = "Anna"`*.*
>
> `=` *stores the value on the right in the variable on the left.*
>
> *Text goes in quotes (*`"Anna"`*); numbers do not (*`17`*).*
>
> `print()` *shows something on the screen:* `print("Hi", name)` *prints* `Hi Anna`*.*

Solve it alone: Create three variables: `name`, `age` and `city`, with your own values. Print one sentence that uses all three, for example: `My name is Anna, I am 17 and I live in Vejle.`

Compare with GenAI:

```
I am a beginner in programming learning the basics of Python.
I only know: variables, assignment (=) and print().

Exercise: Create three variables called name, age and city. Print one sentence that
uses all three, like "My name is Anna, I am 17 and I live in Vejle."

Write a beginner-level solution using only the concepts above. Explain how it works
in 3-4 short sentences.
```

Look for: Did the AI use an f-string (`f"My name is {name}"`)? Ask it what that is.

Exercise 2 — Your age in 5 years

> ***Theory box: data types***
>
> *Every value has a type: `int` (whole number, `17`), `float` (decimal, `1.75`), `str` (text, `"Anna"`), `bool` (`True`/`False`).*
>
> *`type(x)` tells you the type of `x`: `print(type(17))` prints `<class 'int'>`.*
>
> *`"17"` in quotes is text, not a number, so `"17" + 5` gives an error.*
>
> *Convert types with `int("17")` → `17`, `float("1.75")` → `1.75`, `str(17)` → `"17"`.*

Solve it alone: Create four variables: `age = 17`, `height = 1.75`, `name = "Anna"` and `is_student = True`. Print the type of each one. Then create `age_text = "17"` (a number stored as text, as it might come from a web form). Print how old the person will be in 5 years by converting `age_text` to a number first.

Hint: First try `age_text + 5` without converting. What error do you get, and why?

Compare with GenAI:

```
I am a beginner in programming learning the basics of Python.
I only know: variables, assignment (=), print(), and the data types int, float, str
and bool.
From this worksheet I also know: type() and converting types with int(), float() and
str().

Exercise: Create the variables age = 17, height = 1.75, name = "Anna" and is_student
= True, and print the type of each one. Then create age_text = "17" and print how old
the person will be in 5 years by converting age_text to a number.

Write a beginner-level solution using only the concepts above. Explain how it works
in 3-4 short sentences.
```

Look for: Change `age_text` to `"seventeen"` in both programs. What happens? Does the AI warn you about this?

Exercise 3 — Splitting the pizza bill

> ***Theory box: arithmetic operators***
>
> *`+  -  *  /` work as in maths. `/` always gives a `float`: `10 / 4` → `2.5`.*
>
> *`//` is whole-number division (`10 // 4` → `2`); `%` is the remainder (`10 % 4` → `2`).*

> *`round(x, 2)` rounds `x` to 2 decimals.*

Solve it alone: A pizza order costs 389.50 kr and is shared by `friends = 4`. Print how much each person pays, rounded to 2 decimals. Then add a 10% tip to the total and print the new price per person. Change `friends` to 3 and run it again.

Compare with GenAI:

```
I am a beginner in programming learning the basics of Python.
I only know: variables, assignment (=), print(), the data types int, float, str and
bool, and the operators + - * /.
From this worksheet I also know: // (whole division), % (remainder) and round().

Exercise: A pizza order costs 389.50 kr and is shared by friends = 4. Print how much
each person pays, rounded to 2 decimals. Then add a 10% tip to the total and print
the new price per person.

Write a beginner-level solution using only the concepts above. Explain how it works
in 3-4 short sentences.
```

Look for: How did each of you calculate the 10%: `* 1.1`, `* 0.1` then add, or something else? Do you get exactly the same numbers?

Part 2 — Conditionals and loops

Exercise 4 — Even or odd?

> ***Theory box: if / else and comparisons***
>
> *Comparisons give `True` or `False`: `==` equal, `!=` not equal, `<`, `>`, `<=`, `>=`.*
>
> *`if condition:` runs the indented code only when the condition is `True`; `else:` runs otherwise.*
>
> *The colon `:` and the indentation (4 spaces) are required.*
>
> *A number is even when `number % 2 == 0`.*

Solve it alone: Store a whole number in a variable, e.g. `number = 12`. Print `12 is even` or `7 is odd`. Extra: also print `That is a big number!` when it is larger than 1000. Change `number` a few times and run it again to test.

Compare with GenAI:

```
I am a beginner in programming learning the basics of Python.
I only know: variables, assignment (=), print(), the data types int, float, str and
bool, the operators + - * /, comparisons (< > <= >= == !=), and if/else.
From this worksheet I also know: % (remainder).

Exercise: Store a whole number in a variable called number. Print "12 is even" or "7
is odd". Also print "That is a big number!" when it is larger than 1000.

Write a beginner-level solution using only the concepts above. Explain how it works
in 3-4 short sentences.
```

Look for: Is the "big number" check a separate `if`, or inside the `else`? Does it matter? Test with 2000 and 2001.

Exercise 5 — Cinema ticket price

> ***Theory box: elif and combining conditions***
>
> *`elif` ("else if") checks a new condition when the ones above were `False`. Python checks from the top and stops at the first `True`.*
>
> *`and` is `True` only if both sides are; `or` is `True` if at least one side is.*
>
> *`12 <= age < 26` is allowed in Python and means "between 12 and 25".*

Solve it alone: Store an age in a variable, e.g. `age = 17`, and print the ticket price:

Age	Price
Under 12	60 kr
12–25	85 kr
26–64	120 kr
65 and over	70 kr

Extra: add a variable `is_tuesday = True` (or `False`). On Tuesdays every ticket is 20 kr cheaper.

Compare with GenAI:

```
I am a beginner in programming learning the basics of Python.
I only know: variables, assignment (=), print(), the data types int, float, str and
bool, the operators + - * /, comparisons (< > <= >= == !=), and/or, and if/else.
From this worksheet I also know: elif.

Exercise: Store an age in a variable called age and print the cinema ticket price:
under 12 = 60 kr, 12-25 = 85 kr, 26-64 = 120 kr, 65 and over = 70 kr. Add a variable
is_tuesday (True or False). On Tuesdays every ticket is 20 kr cheaper.

Write a beginner-level solution using only the concepts above. Explain how it works
in 3-4 short sentences.
```

Look for: Test the edge cases 11, 12, 25, 26, 64 and 65 in both programs. How does each of you check the day: `if is_tuesday:` or `if is_tuesday == True:`? Do both work?

Exercise 6 — Times table

> ***Theory box: for loops***
>
> *A `for` loop repeats code once for each value: `for i in range(1, 4):` gives `i` = 1, 2, 3.*
>
> *`range(start, stop)` stops **before** `stop`. `range(5)` gives 0, 1, 2, 3, 4.*
>
> *Everything indented under the `for` line is repeated.*

Solve it alone: Store a number in a variable, e.g. `number = 7`, and print its times table from 1 to 10, like `7 x 3 = 21`. Extra: after the table, print the sum of all ten results.

Compare with GenAI:

```
I am a beginner in programming learning the basics of Python.
I only know: variables, assignment (=), print(), the data types int, float, str, bool
and list, the operators + - * /, comparisons, and/or, if/else, and for loops over a
list.
From this worksheet I also know: elif and range().

Exercise: Store a number in a variable called number and print its times table from 1
to 10, like "7 x 3 = 21". After the table, print the sum of all ten results.

Write a beginner-level solution using only the concepts above. Explain how it works
in 3-4 short sentences.
```

Look for: Where does each program create the sum variable, inside or before the loop? Move it inside the loop in your code and see what breaks.

Part 3 — Loops with conditions, and functions

Exercise 7 — Saving up

> ***Theory box: while loops***
>
> *A `while` loop repeats as long as its condition is `True`: `while total < 1000:`.*
>
> *Something inside the loop must change, or the loop never stops.*
>
> *A counter is a variable you increase each round: `weeks = weeks + 1`.*

Solve it alone: You save 50 kr in week 1. Every week after that you save 10 kr more than the week before (60 kr, 70 kr, …). Use a `while` loop to find out how many weeks it takes until you have saved at least 1000 kr. Print the number of weeks and the total saved.

Compare with GenAI:

```
I am a beginner in programming learning the basics of Python.
I only know: variables, assignment (=), print(), the data types int, float, str, bool
and list, the operators + - * /, comparisons, and/or, if/else, for loops and while
loops.
From this worksheet I also know: elif.

Exercise: You save 50 kr in week 1. Every week after that you save 10 kr more than
the week before (60 kr, 70 kr, ...). Use a while loop to find how many weeks it takes
until you have saved at least 1000 kr. Print the number of weeks and the total saved.

Write a beginner-level solution using only the concepts above. Explain how it works
in 3-4 short sentences.
```

Look for: Work out the first three weeks by hand (50, 110, 180 kr). Do you and the AI get the same number of weeks, or is one of you off by one? Did the AI use `while True` or `break`? Those were not on the list.

Exercise 8 — A week of weather in Vejle

> ***Theory box: lists and looping over them***
>
> *A list stores many values in order:* `temps = [8, 11, 7]`*.*
>
> `temps[0]` *is the first value;* `len(temps)` *is the number of values.*
>
> `for t in temps:` *runs once for each value in the list.*
>
> *To find a total or a maximum, start with a variable before the loop and update it inside the loop.*

Solve it alone: Use this list of daily temperatures (°C):

```
temps = [9, 12, 7, 14, 11, 6, 10]
```

Print the average temperature, the warmest temperature, and how many days were below 10 °C. Rule: do **not** use `sum()`, `max()` or `min()`. Find them with a loop.

Compare with GenAI:

```
I am a beginner in programming learning the basics of Python.
I only know: variables, assignment (=), print(), the data types int, float, str, bool
and list, the operators + - * /, comparisons, and/or, if/else, for and while loops,
and lists with append(), remove() and len().
From this worksheet I also know: elif.

Exercise: Given temps = [9, 12, 7, 14, 11, 6, 10], print the average temperature, the
warmest temperature, and how many days were below 10 degrees. Do not use sum(), max()
or min(); use loops.

Write a beginner-level solution using only the concepts above. Explain how it works
in 3-4 short sentences.
```

Look for: Did the AI follow the "no `sum()`/`max()`" rule? What starting value did each of you use for the warmest temperature? Would it still work if every day were below 0 °C?

Exercise 9 — Is the password strong?

> ***Theory box: functions***
>
> `def name(parameter):` *defines a function: a reusable piece of code with a name.*
>
> *The parameter is the input;* `return` *sends a result back out.*
>
> *The code inside only runs when you call it:* `result = name("abc")`*.*
>
> *You can loop over the characters of a text:* `for ch in "abc":`*.* `ch.isdigit()` *is* `True` *if* `ch` *is a digit.*

Solve it alone: Write a function `is_strong(password)` that returns `True` if the password has at least 8 characters **and** contains at least one digit, otherwise `False`. Test it:

```
print(is_strong("hello"))       # False
print(is_strong("helloworld"))  # False
print(is_strong("hello2026"))   # True
```

Compare with GenAI:

```
I am a beginner in programming learning the basics of Python.
I only know: variables, assignment (=), print(), the data types int, float, str, bool
and list, the operators + - * /, comparisons, and/or, if/else, for and while loops,
lists with append(), remove() and len(), and functions with def, parameters and
return values.
From this worksheet I also know: elif, looping over the characters of a string, and
isdigit().

Exercise: Write a function is_strong(password) that returns True if the password has
at least 8 characters and contains at least one digit, otherwise False. Test it with
"hello", "helloworld" and "hello2026".

Write a beginner-level solution using only the concepts above. Explain how it works
in 3-4 short sentences.
```

Look for: Does either function `print` instead of `return`? Then ask the AI: "Add one more rule: the password must also contain an uppercase letter." How much of the code had to change?

Exercise 10 — Build a mini chatbot

> ***Theory box: putting it all together***
>
> *A program can use functions, loops and conditions together: a loop goes through the messages, a function decides each answer.*
>
> *`word in text` is `True` if the word appears in the text: `"hi" in "hi there"` → `True`.*
>
> *`text.lower()` turns text into lowercase, so `"Hi"` and `"hi"` are treated the same.*

Solve it alone: Write a function `reply(message)` that returns an answer:

- If the message is `bye` → `Goodbye!`
- If it contains `hello` or `hi` → `Hello! What do you want to study?`
- If it contains `python` → `Python is a great first language!`
- If it contains `ai` → `AI is all about learning from data.`
- Otherwise → `Sorry, I don't understand. Try another question.`

Then test your function on this list. Loop over it and print each message followed by the chatbot's reply:

```
messages = ["Hi there", "I like Python", "Tell me about AI", "I like trains", "bye"]
```

Compare with GenAI:

```
I am a beginner in programming learning the basics of Python.
I only know: variables, assignment (=), print(), the data types int, float, str, bool
and list, the operators + - * /, comparisons, and/or, if/else, for and while loops,
lists with append(), remove() and len(), and functions with def, parameters and
return values.
From this worksheet I also know: elif, the in operator and lower().

Exercise: Write a function reply(message) that returns an answer: if the message is
"bye" return "Goodbye!"; if it contains "hello" or "hi" return "Hello! What do you
want to study?"; if it contains "python" return "Python is a great first language!";
if it contains "ai" return "AI is all about learning from data."; otherwise return
```

```
"Sorry, I don't understand. Try another question." Then loop over the list messages = ["Hi there", "I like Python", "Tell me about AI", "I like trains", "bye"] and print each message followed by the reply.

Write a beginner-level solution using only the concepts above. Explain how it works in 3-4 short sentences.
```

Look for: Type `I like trains` into both chatbots. Why might it answer about AI? (Hint: look for `ai` inside `trains`.) Does the AI's chatbot have the same bug? Your chatbot follows fixed rules; the chatbot you are talking to learned from data. What difference do you notice?

Final reflection

Answer in one or two sentences each, alone or with the person next to you.

1. In which exercise was your solution closest to the AI's? Where was it most different?
2. Did the AI ever use something you had not learned, ignore a rule, or make a mistake? Which exercise?
3. Was it easier to understand the AI's code because you had tried the exercise yourself first?
4. When would you use GenAI while learning to program, and when would you rather not?

Tip: the more precisely you describe the problem and what you know, the better the AI's answer. That is the **same skill as programming itself: telling a computer exactly what you want.**